

Einstein/CTensor

Setup

```
In[*]:= << mGRG`Einstein`
```

```
In[*]:= $PreRead = ReplaceAll[#,  
    expr_String -> StringReplace[expr, "EP" -> "mGRG`Einstein`Private"]] &;
```

InitCTensor (Set/ClearCTensor)

Initialization of Component Tensors in Coordinate Basis

```
In[*]:= SetCTensor::usage
```

Out[*]=

SetCTensor is an alias for InitCTensor.

```
In[*]:= InitCTensor::usage
```

Out[*]=

InitCTensor[coSys, metric, <basis>, opts] initializes the environment for component tensor calculations. `coSys` is a list of coordinates, and `metric` is the metric matrix. An optional `basis` matrix can be provided for non-coordinate bases. Options allow for pre-calculation of curvature components and setting the simplification level.

```
In[*]:= SimplifyMore::usage
```

Out[*]=

SimplifyMore is an option for InitCTensor. If set to True, it uses CsimpifyMore as the default simplification method for calculations.

```
In[*]:= Verbose::usage
```

Out[*]=

Verbose is an option for functions like Tsimpify and InitCTensor. If set to True, it prints detailed information about the calculation steps or timing.

In[*]:= **FourDimensionQ::usage**

Out[*]=

FourDimensionQ is an option for InitCTensor. If True (the default), it enforces that the spacetime is four-dimensional. Set to False to allow for calculations in other dimensions.

좌표계와 계량 텐서를 입력으로 받아 Coordinate Basis에서 성분 텐서를 초기화 한다. 옵션으로 SimplifyMore, Verbose, GammaCD, RiemannCD, RicciCD, FourDimensionQ가 있다.

In[*]:= **coSys = {t, x};
metric = {{-e^{2ρ[t,x]}, 0}, {0, e^{2ρ[t,x]}}}; MatrixForm[metric]**

Out[*]//MatrixForm=

$$\begin{pmatrix} -e^{2\rho[t,x]} & 0 \\ 0 & e^{2\rho[t,x]} \end{pmatrix}$$

In[*]:= **InitCTensor[coSys, metric, RicciCD → True]**

Msg: It is assumed a four-dimensional spacetime. But the number of coordinates is not four!
Use FourDimensionQ → False option

Out[*]=

\$Failed

In[*]:= **InitCTensor[coSys, metric, RicciCD → True, FourDimensionQ → False];
Show[DefaultKind]**

Out[*]//TableForm=

AutoFlag	True
MarkErrorFlag	True
ResetDummiesFlag	True
SyntaxCheckFlag	False
KdeltaFlag	True
MetricgFlag	True
InitCTensorFlag	True
TorsionFreeQ of CD	True
-----	-----
Kind	Latin
Dimension	2
Sig	Any
Coordinates	t x
CoordinateBasisQ	True
EvaluateBDFlag	True

In[*]:= **Show[LineElement]**

Out[*]=

$$ds^2 = -e^{2\rho[t,x]} (dt^2 - dx^2)$$

In[*]:= **Show[Metricg]**

Symmetry: $g_{(ab)}$

$$g_{tt} = -e^{2\rho[t,x]}$$

$$g_{xx} = e^{2\rho[t,x]}$$

In[*]:= Show[Structuref]

$$f_{ab}{}^c = 0$$

In[*]:= Show[GammaCD]

Symmetry: $\Gamma_{(ab)}{}^c$

$$\Gamma_{tt}{}^t = \rho^{(1,0)}[t, x]$$

$$\Gamma_{tt}{}^x = \rho^{(0,1)}[t, x]$$

$$\Gamma_{tx}{}^t = \rho^{(0,1)}[t, x]$$

$$\Gamma_{tx}{}^x = \rho^{(1,0)}[t, x]$$

$$\Gamma_{xx}{}^t = \rho^{(1,0)}[t, x]$$

$$\Gamma_{xx}{}^x = \rho^{(0,1)}[t, x]$$

In[*]:= Show[RiemannCD]

Symmetry: $R_{[ab][cd]} = R_{[cd][ab]}$

$$R_{txtx} = e^{2\rho[t,x]} (\rho^{(0,2)}[t, x] - \rho^{(2,0)}[t, x])$$

In[*]:= Show[RicciCD]

Symmetry: $R_{(ab)}$

$$R_{tt} = \rho^{(0,2)}[t, x] - \rho^{(2,0)}[t, x]$$

$$R_{xx} = -\rho^{(0,2)}[t, x] + \rho^{(2,0)}[t, x]$$

In[*]:= Show[ScalarCD]

$$R = -2 e^{-2\rho[t,x]} (\rho^{(0,2)}[t, x] - \rho^{(2,0)}[t, x])$$

Clear Component Tensors in Coordinate Basis

In[*]:= ClearCTensor::usage

Out[*]=

ClearCTensor[] clears all previously calculated tensor components and resets the CTensor environment.

```
In[*]:= ClearCTensor[];
Show[DefaultKind]
```

```
Out[*]//TableForm=
```

AutoFlag	True
MarkErrorFlag	True
ResetDummiesFlag	True
SyntaxCheckFlag	False
KdeltaFlag	True
MetricgFlag	True
InitCTensorFlag	False
TorsionFreeQ of CD	True
-----	-----
Kind	Latin
Dimension	Any
Sig	Any
Coordinates	none
CoordinateBasisQ	True
EvaluateBDFlag	False

```
In[*]:= {Metricg[-1, -1], Metricg[1, 1]}
```

```
Out[*]=
```

```
{g11, g11}
```

```
In[*]:= EP`bdMetric[{1, 1, 1}]
```

```
Out[*]=
```

```
mGRG`Einstein`Private`bdMetric[{1, 1, 1}]
```

```
In[*]:= {Structuref[-1, -2, -1], Structuref[-1, -2, 1]}
```

```
Out[*]=
```

```
{Structuref[-1, -2, -1], Structuref[-1, -2, 1]}
```

```
In[*]:= {GammaCD[-1, -1, -1], GammaCD[-1, -1, 1]}
```

```
Out[*]=
```

```
{Γ111, Γ111}
```

```
In[*]:= {RiemannCD[-1, -2, -1, -2], RicciCD[-1, -1], RicciCD[1, 1], ScalarCD[]}
```

```
Out[*]=
```

```
{R1212, R11, R11, R}
```

```
In[*]:= {EP`inverseBasisMatrix, mGRG`STensor`Private`basisMatrix[DefaultKind]}
```

```
Out[*]=
```

```
{mGRG`Einstein`Private`inverseBasisMatrix,
mGRG`STensor`Private`basisMatrix[Latin]}
```

```
In[*]:= {EP`cMetric, EP`nDimension}
```

```
Out[*]=
```

```
{mGRG`Einstein`Private`cMetric, mGRG`Einstein`Private`nDimension}
```

```
In[*]:= {mGRG`STensor`Private`flagTable[InitCTensorFlag],  
mGRG`STensor`Private`flagTable[EvaluateBDFlag[DefaultKind]]}
```

```
Out[*]=
```

```
{False, False}
```

```
In[*]:= {EP`defaultCTensor[Verbose], EP`defaultCTensor[EP`simpMethod]}
```

```
Out[*]=
```

```
{mGRG`Einstein`Private`defaultCTensor[Verbose],  
mGRG`Einstein`Private`defaultCTensor[mGRG`Einstein`Private`simpMethod]}
```

Initialization of Component Tensor in Non-Coordinate Basis

좌표계, metric, 기준 벡터의 행렬 표현을 입력으로 받아 Non-coordinate Basis에서 성분 텐서를 초기화한다.

```
In[*]:= coSys = {t, x};
```

Orthonormal Basis ($e_a^A = h_a^\mu \partial_\mu^A$), where basisM = h_a^μ :

```
In[*]:= metric = {{-1, 0}, {0, 1}};  
basisM = {{e-ρ[t,x], 0}, {0, e-ρ[t,x]}}; MatrixForm[basisM]
```

```
Out[*]//MatrixForm=
```

$$\begin{pmatrix} e^{-\rho[t,x]} & 0 \\ 0 & e^{-\rho[t,x]} \end{pmatrix}$$

```
In[*]:= InitCTensor[coSys, metric, basisM, RicciCD → True, FourDimensionQ → False];
Show[DefaultKind]
```

```
Out[*]//TableForm=
```

AutoFlag	True
MarkErrorFlag	True
ResetDummiesFlag	True
SyntaxCheckFlag	False
KdeltaFlag	True
MetricgFlag	True
InitCTensorFlag	True
TorsionFreeQ of CD	True
-----	-----
Kind	Latin
Dimension	2
Sig	Any
Coordinates	t x
CoordinateBasisQ	False
EvaluateBDFlag	True

```
In[*]:= Show[LineElement]
```

```
Out[*]=
```

$$ds^2 = -dt^2 + dx^2$$

```
In[*]:= Show[Metricg]
```

Symmetry: $g_{(ab)}$

$$g_{11} = -1$$

$$g_{22} = 1$$

```
In[*]:= Show[Structuref]
```

Symmetry: $f_{[ab]}^c$

$$f_{12}^1 = e^{-\rho[t,x]} \rho^{(0,1)}[t, x]$$

$$f_{12}^2 = -e^{-\rho[t,x]} \rho^{(1,0)}[t, x]$$

```
In[*]:= Show[GammaCD]
```

$$\Gamma_{11}^2 = e^{-\rho[t,x]} \rho^{(0,1)}[t, x]$$

$$\Gamma_{12}^1 = e^{-\rho[t,x]} \rho^{(0,1)}[t, x]$$

$$\Gamma_{21}^2 = e^{-\rho[t,x]} \rho^{(1,0)}[t, x]$$

$$\Gamma_{22}^1 = e^{-\rho[t,x]} \rho^{(1,0)}[t, x]$$

```
In[*]:= Show[RiemannCD]
```

Symmetry: $R_{[ab][cd]} = R_{[cd][ab]}$

$$R_{1212} = e^{-2\rho[t,x]} (\rho^{(0,2)}[t, x] - \rho^{(2,0)}[t, x])$$

```
In[*]:= Show[RicciCD]
```

Symmetry: $R_{(ab)}$

$$R_{11} = e^{-2\rho[t,x]} (\rho^{(0,2)}[t,x] - \rho^{(2,0)}[t,x])$$

$$R_{22} = -e^{-2\rho[t,x]} (\rho^{(0,2)}[t,x] - \rho^{(2,0)}[t,x])$$

In[*]:= Show[ScalarCD]

$$R = -2 e^{-2\rho[t,x]} (\rho^{(0,2)}[t,x] - \rho^{(2,0)}[t,x])$$

Clear Component Tensors in Non-Coordinate Basis

In[*]:= ClearCTensor[];
Show[DefaultKind]

Out[*]//TableForm=

AutoFlag	True
MarkErrorFlag	True
ResetDummiesFlag	True
SyntaxCheckFlag	False
KdeltaFlag	True
MetricgFlag	True
InitCTensorFlag	False
TorsionFreeQ of CD	True
-----	-----
Kind	Latin
Dimension	Any
Sig	Any
Coordinates	none
CoordinateBasisQ	False
EvaluateBDFlag	False

In[*]:= {Metricg[-1, -1], Metricg[1, 1]}

Out[*]=

{ g_{11} , g^{11} }

In[*]:= EP`bdMetric[{1, 1, 1}]

Out[*]=

mGRG`Einstein`Private`bdMetric[{1, 1, 1}]

In[*]:= {Structuref[-1, -2, -1], Structuref[-1, -2, 1]}

Out[*]=

{ f_{121} , f_{12}^1 }

In[*]:= {GammaCD[-1, -1, -1], GammaCD[-1, -1, 1]}

Out[*]=

{ Γ_{111} , Γ_{11}^1 }

```
In[*]:= {RiemannCD[-1, -2, -1, -2], RicciCD[-1, -1], RicciCD[1, 1], ScalarCD[]}
```

```
Out[*]= {R1212, R11, R11, R}
```

```
In[*]:= {EP`inverseBasisMatrix, mGRG`STensor`Private`basisMatrix[DefaultKind]}
```

```
Out[*]= {mGRG`Einstein`Private`inverseBasisMatrix,
mGRG`STensor`Private`basisMatrix[Latin]}
```

```
In[*]:= {EP`cMetric, EP`nDimension}
```

```
Out[*]= {mGRG`Einstein`Private`cMetric, mGRG`Einstein`Private`nDimension}
```

```
In[*]:= {mGRG`STensor`Private`flagTable[InitCTensorFlag],
mGRG`STensor`Private`flagTable[EvaluateBDFlag[DefaultKind]]}
```

```
Out[*]= {False, False}
```

```
In[*]:= {EP`defaultCTensor[Verbose], EP`defaultCTensor[EP`simplMethod]}
```

```
Out[*]= {mGRG`Einstein`Private`defaultCTensor[Verbose],
mGRG`Einstein`Private`defaultCTensor[mGRG`Einstein`Private`simplMethod]}
```

InitCTensor from a File

좌표계, metric, InitCTensor 명령, 옵션 등을 포함한 파일이 MetricPath/filename.m에 있을 때, filename을 입력으로 받아 성분 텐서를 초기화한다. 여기서 MetricPath == mGRG`Einstein`metrics`이다. 다른 위치의 파일을 사용하려면 MetricPath의 값을 변경한다.

```
In[*]:= MetricPath
```

```
Out[*]= mGRG`Einstein`Metrics`
```

```
In[*]:= Timing[InitCTensor["Schwarzschild"]]
```

Coordinate System = {t, r, θ , ϕ }

```
Out[*]= {0.042113, Null}
```

```
In[*]:= Show[LineElement, Simplify]
```

```
Out[*]=
```

$$ds^2 = \frac{dr^2}{1 - \frac{2GM}{r}} + \left(-1 + \frac{2GM}{r}\right) dt^2 + r^2 (d\theta^2 + d\phi^2 \sin[\theta]^2)$$

In[*]:=

Show[Metricg]Symmetry: $g_{(ab)}$

$$g_{tt} = -1 + \frac{2GM}{r}$$

$$g_{rr} = \frac{1}{1 - \frac{2GM}{r}}$$

$$g_{\theta\theta} = r^2$$

$$g_{\phi\phi} = r^2 \sin[\theta]^2$$

In[*]:=

Show[Structuref]

$$f_{ab}{}^c = 0$$

In[*]:=

Show[GammaCD]Symmetry: $\Gamma_{(ab)}{}^c$

$$\Gamma_{tt}{}^r = \frac{-2G^2M^2 + GM r}{r^3}$$

$$\Gamma_{tr}{}^t = -\frac{GM}{(2GM - r)r}$$

$$\Gamma_{rr}{}^r = \frac{GM}{(2GM - r)r}$$

$$\Gamma_{r\theta}{}^\theta = \frac{1}{r}$$

$$\Gamma_{r\phi}{}^\phi = \frac{1}{r}$$

$$\Gamma_{\theta\theta}{}^r = 2GM - r$$

$$\Gamma_{\theta\phi}{}^\phi = \cot[\theta]$$

$$\Gamma_{\phi\phi}{}^r = \frac{1}{2} (2GM - r - 2GM \cos[2\theta] + r \cos[2\theta])$$

$$\Gamma_{\phi\phi}{}^\theta = -\frac{1}{2} \sin[2\theta]$$

In[*]:=

Show[RiemannCD]Symmetry: $R_{[ab][cd]} = R_{[cd][ab]}$

$$R_{trtr} = -\frac{2GM}{r^3}$$

$$R_{t\theta t\theta} = \frac{-2G^2M^2 + GM r}{r^2}$$

$$R_{t\phi t\phi} = \frac{-2G^2M^2 + GM r + 2G^2M^2 \cos[2\theta] - GM r \cos[2\theta]}{2r^2}$$

$$R_{r\theta r\theta} = \frac{GM}{2GM - r}$$

$$R_{r\phi r\phi} = \frac{GM - GM \cos[2\theta]}{2(2GM - r)}$$

$$R_{\theta\phi\theta\phi} = GM r - GM r \cos[2\theta]$$

```
In[*]:= Show[RicciCD]
```

Symmetry: $R_{(ab)}$

$R_{ab} = 0$

```
In[*]:= Show[ScalarCD]
```

$R = 0$

Clear Component Tensors

```
In[*]:= ClearCTensor[]
```

```
Show[DefaultKind]
```

```
Out[*]//TableForm=
```

AutoFlag	True
MarkErrorFlag	True
ResetDummiesFlag	True
SyntaxCheckFlag	False
KdeltaFlag	True
MetricgFlag	True
InitCTensorFlag	False
TorsionFreeQ of CD	True
-----	-----
Kind	Latin
Dimension	Any
Sig	Any
Coordinates	none
CoordinateBasisQ	True
EvaluateBDFlag	False

Calculations of Component Tensors

Coordinates, Metric, and InitCTensor

```
In[*]:= Tcalc::usage
```

```
Out[*]=
```

Tcalc[tensor, simpCmd] calculates the components of a given `tensor` (e.g., GammaCD, RicciCD, RiemannCD). An optional simplification command `simpCmd` can be specified.

Tcalc 함수의 인자로 Structuref, GammaCD, RiemannCD, RicciCD, ScalarCD 등을 사용하여 f_{ab}^c , Γ_{ab}^c , R_{abc}^d , R_{ab} , R 을 각각 계산한다. 옵션으로는 simplification을 위한 함수의 이름이 될 수 있다. 이 함수를 사용하려면 InitCTensor 함수를 이용하여 성분 텐서를 먼저 초기화시켜야 한다.

Coordinate Systems, Options, and Metric for the Schwarzschild SpaceTime:

```
In[*]:= coSys = {t, r,  $\theta$ ,  $\phi$ };
SetOptions[Together, Trig  $\rightarrow$  True]; (* for default Csimpify *)
metric = {{- $\left(1 - \frac{2GM}{r}\right)$ , 0, 0, 0},
          {0,  $\frac{1}{1 - \frac{2GM}{r}}$ , 0, 0}, {0, 0,  $r^2$ , 0}, {0, 0, 0,  $r^2 \sin[\theta]^2$ }};
```

```
In[*]:= InitCTensor[coSys, metric, Verbose  $\rightarrow$  True]
```

Total setup time: 0.s

Structure constant

```
In[*]:= Tcalc[Structuref]
```

```
In[*]:= Show[Structuref]
```

$$f_{ab}^c = 0$$

Metric connection

```
In[*]:= Tcalc[GammaCD]
```

Calculated g^{ab} using Csimpify in 0.01s

Calculated $\partial_a g_{bc}$ using Csimpify in 0.s

Calculated Γ_{abc} using Csimpify in 0.s

Calculated Γ_{ab}^c using Csimpify in 0.s

```
In[*]:= Show[GammaCD]
```

Symmetry: $\Gamma_{(ab)}^c$

$$\Gamma_{tt}^r = \frac{-2G^2M^2 + GMr}{r^3}$$

$$\Gamma_{tr}^t = -\frac{GM}{(2GM - r)r}$$

$$\Gamma_{rr}^r = \frac{GM}{(2GM - r)r}$$

$$\Gamma_{r\theta}^\theta = \frac{1}{r}$$

$$\Gamma_{r\phi}^\phi = \frac{1}{r}$$

$$\Gamma_{\theta\theta}^r = 2GM - r$$

$$\Gamma_{\theta\phi}^\phi = \cot[\theta]$$

$$\Gamma_{\phi\phi}^r = \frac{1}{2} (2GM - r - 2GM \cos[2\theta] + r \cos[2\theta])$$

$$\Gamma_{\phi\phi}^\theta = -\frac{1}{2} \sin[2\theta]$$

Curvature tensors

In[*]:= Tcalc[RicciCD]

Calculated R_{abcd} using Csimpify in 0.01s

Calculated R_{ab} using Csimpify in 0.01s

In[*]:= Show[RicciCD]

Symmetry: $R_{(ab)}$

$R_{ab} = 0$

In[*]:= Tcalc[ScalarCD]

Calculated R using Csimpify in 0.s

In[*]:= Show[ScalarCD]

$R = 0$

In[*]:= Tcalc[RiemannCD[-1, -2, -1, -2]]; RiemannCD[-1, -2, -1, -2]

Calculated R_{trtr} using Csimpify in 0.s

Out[*]=

$$-\frac{2 \text{ G M}}{r^3}$$

In[*]:= Tcalc[RiemannCD, Simplify]

Calculated R_{abcd} using Simplify in 0.02s

In[*]:= Show[RiemannCD]

Symmetry: $R_{[ab][cd]} = R_{[cd][ab]}$

$$R_{trtr} = -\frac{2 \text{ G M}}{r^3}$$

$$R_{t\theta t\theta} = -\frac{\text{G M} (2 \text{ G M} - r)}{r^2}$$

$$R_{t\phi t\phi} = -\frac{\text{G M} (2 \text{ G M} - r) \sin[\theta]^2}{r^2}$$

$$R_{r\theta r\theta} = \frac{\text{G M}}{2 \text{ G M} - r}$$

$$R_{r\phi r\phi} = \frac{\text{G M} \sin[\theta]^2}{2 \text{ G M} - r}$$

$$R_{\theta\phi\theta\phi} = 2 \text{ G M } r \sin[\theta]^2$$

Clear Component Tensors

```
In[*]:= ClearCTensor[]
Show[DefaultKind]
```

Out[*]//TableForm=

AutoFlag	True
MarkErrorFlag	True
ResetDummiesFlag	True
SyntaxCheckFlag	False
KdeltaFlag	True
MetricgFlag	True
InitCTensorFlag	False
TorsionFreeQ of CD	True
-----	-----
Kind	Latin
Dimension	Any
Sig	Any
Coordinates	none
CoordinateBasisQ	True
EvaluateBDFlag	False

Commands in CTensor

Geodesic

```
In[*]:= Geodesic::usage
```

Out[*]=

Geodesic[comp, simpCmd] calculates the geodesic equation for the specified component `comp`. It is only available for coordinate bases. `simpCmd` is an optional simplification function.

시공간의 geodesic을 얻는다. 인자는 정수로 좌표계의 순서이다. 옵션으로 simplification을 위한 함수가 될 수 있다. 이 함수를 사용하려면 InitCTensor 함수를 이용하여 성분 텐서를 초기화하여야 한다.

```
In[*]:= coSys = {t, r, θ, ϕ};
metric = {{-(1 - 2GM/r), 0, 0, 0},
          {0, 1/(1 - 2GM/r), 0, 0}, {0, 0, r^2, 0}, {0, 0, 0, r^2 Sin[θ]^2}};
```

```
In[*]:= SetCTensor[coSys, metric, GammaCD → True]
```

```
In[*]:= Geodesic[1, Simplify]
```

Out[*]=

$$-\frac{2GM\dot{r}\ddot{t}}{2GMr - r^2} + \ddot{t}$$

In[*]:= **Geodesic[2, Simplify]**

Out[*]=

$$\frac{G M \dot{r}^2}{2 G M r - r^2} + \frac{G M (-2 G M + r) \dot{t}^2}{r^3} + (2 G M - r) \dot{\theta}^2 + \ddot{r} + (2 G M - r) \dot{\phi}^2 \sin[\theta]^2$$

In[*]:= **Geodesic[3, Simplify]**

Out[*]=

$$\frac{2 \dot{r} \dot{\theta}}{r} + \ddot{\theta} - \cos[\theta] \dot{\phi}^2 \sin[\theta]$$

In[*]:= **Geodesic[4, Simplify]**

Out[*]=

$$\frac{2 \dot{r} \dot{\phi}}{r} + 2 \cot[\theta] \dot{\theta} \dot{\phi} + \ddot{\phi}$$

Clear Component Tensors

In[*]:= **ClearCTensor[]**